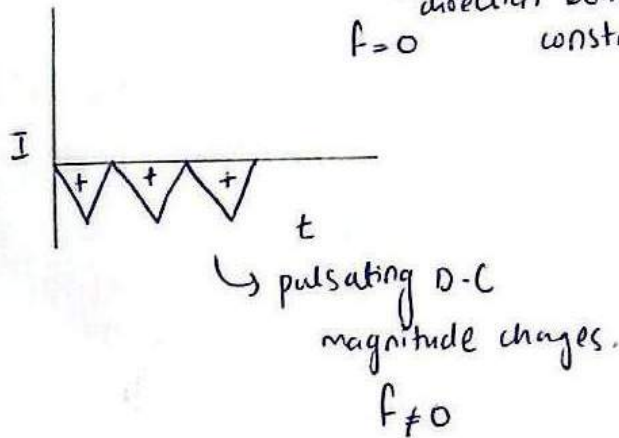
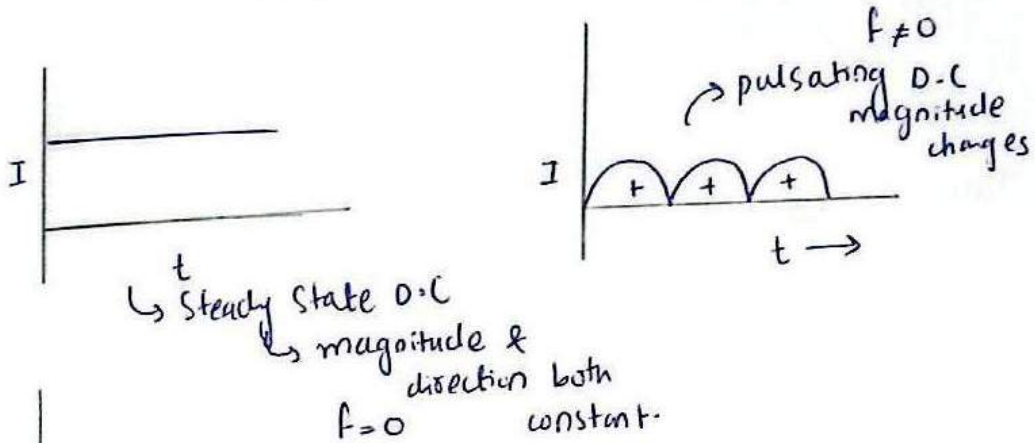
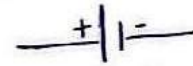


Source: AC generator Alternating Current:

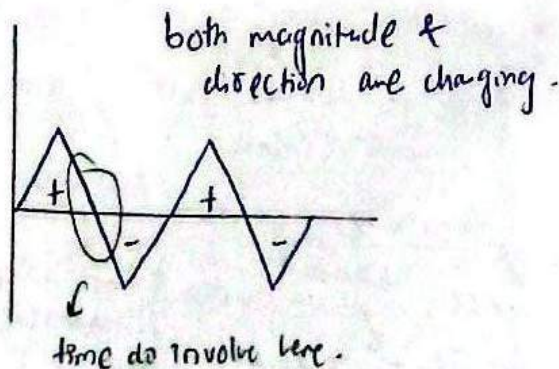
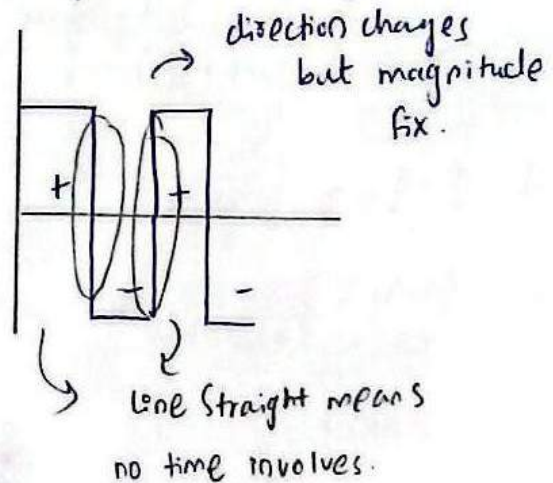
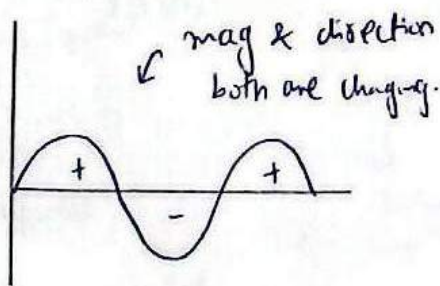
Direct current $\Rightarrow$ unidirectional current, it flows only in one direction.



Symbol of Source



Alternating current $\Rightarrow$ direction change continuously.



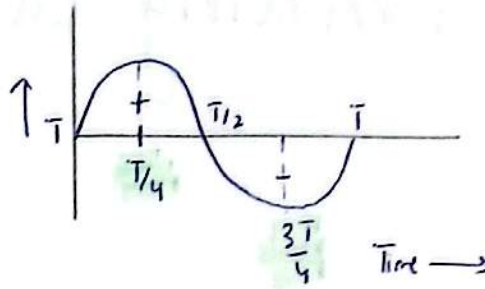
Sinusoidal A.C

$$I = I_0 \sin \omega t$$

$$V = V_0 \sin \omega t$$

or

$$E = E_0 \sin \omega t$$

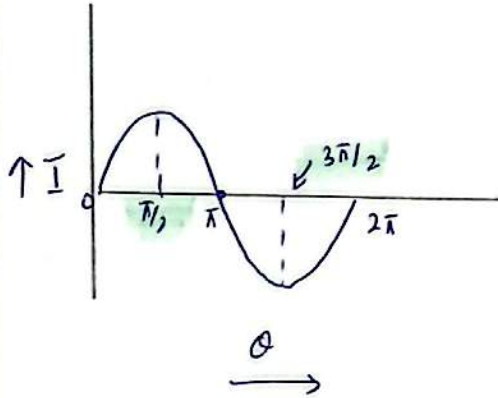


the peak $\Rightarrow T/4$

the peak $\Rightarrow 3T/4$

Zero value $(0, T/2, T)$

$(0, \pi, 2\pi)$



$$I = I_0 \sin \theta$$

$$I = I_0 \sin \omega t$$

$$I = I_0 \sin \left(\frac{2\pi}{T} \right) t$$

$$I = I_0 \sin (2\pi f) t$$

At what time AC becomes half of its maximum value.

$$I = \frac{1}{2} I_0$$

$$\sin \theta = \frac{1}{2}$$

$$\theta = 30^\circ = \frac{\pi}{6} \text{ rad.}$$

$$t = \frac{T}{6 \times 2} = \frac{T}{12}$$

$$I = I_0 \sin (\omega t + \phi)$$

initial phase.

$$V = V_0 \sin (\omega t + \phi)$$

if $\phi = 0$ $I = I_0 \sin \omega t$

$\phi = 90^\circ$ $I = I_0 \cos \omega t$

$\phi = 180^\circ$ $I = -I_0 \sin \omega t$

$\phi = 270^\circ$ $I = -I_0 \cos \omega t$

(Nature)

current

D.C

Scalar

not obeys laws of vector addition.

* currents obeys KCL

add & subtract just like scalars.

Measure

common Ammeter

For A.C $I = 0$

I_{av}

common Voltmeter

For A.C $V = 0$

V_{av}

A.C

① phasor

Anti-clockwise rotating vector.

③ vector

Measure.

Ammeter

hot wire ammeter

I_{rms}

voltmeter

hot wire voltmeter

V_{rms}

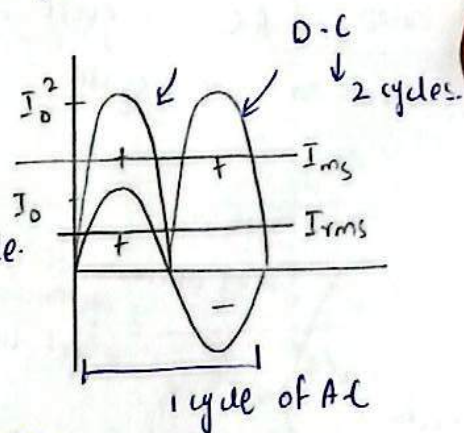
- ↳ hot wire ammeters can measure both AC & DC
 ↳ hot wire ammeters have non-uniform scale as expansion occurs which is not in uniform way.
 ↳ based on principle of thermal expansion & its non-uniform process.

method not for sinusoidal wave form but as in book it is in this way....
 Average value of A.C for one complete cycle = $I_{av} = 0$
 $I_{av} = \frac{+I_0 - I_0}{2} = 0$ (max value → min value)
 $V_{av} = 0 \rightarrow 1 \text{ complete cycle.}$

$I_{av} = \frac{2I_0}{\pi}$
 half cycle.

$V_{av} = \frac{2V_0}{\pi}$
 half cycle

unit Ampere²
 $I_{rms}^2 = \frac{0 + I_0^2}{2} = \frac{I_0^2}{2}$
 mean square
 ↓
 (2) (1)
 complete cycle.



$\sqrt{I_{rms}^2} = \frac{\sqrt{I_0^2}}{\sqrt{2}} = \frac{I_0}{\sqrt{2}}$ unit Ampere.
 ↓
 Root mean Square

if $I_0 = 3A$
 $I_0^2 = 9A^2$

$\frac{I_0^2}{2} = \frac{9}{2} = 4.5A^2$

$I_{rms} = \frac{I_0}{\sqrt{2}}$

$V_{rms} = \frac{V_0}{\sqrt{2}}$

$I_{rms} = 4.5$

$\sqrt{I_{rms}^2} = \sqrt{4.5}$

↓
 aur kam
 hu jaye gi

$I_{rms} = 0.707 I_0$

$V_{rms} = 0.707 V_0$

$I_0 = \sqrt{2} I_{rms}$

$V_0 = \sqrt{2} V_{rms}$

$I, I_0, I_{av}, I_{ms}, I_{rms}$

↓ by
↳ measure Ammeter.

↳ effective value or measurable value.
↳ equivalent to D.C

A.C circuit

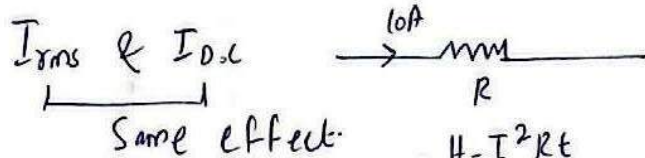
$$I = 10A$$

↳ ? if not mention.

↓
then take as
root mean square

$$V = 220V_{rms}$$

↓
root mean square value of voltage.



$$H = I^2 R t$$

Same heat produced by both.
 $I_{D.C}$ & I_{rms} .

$$I = I_0 \sin \omega t$$

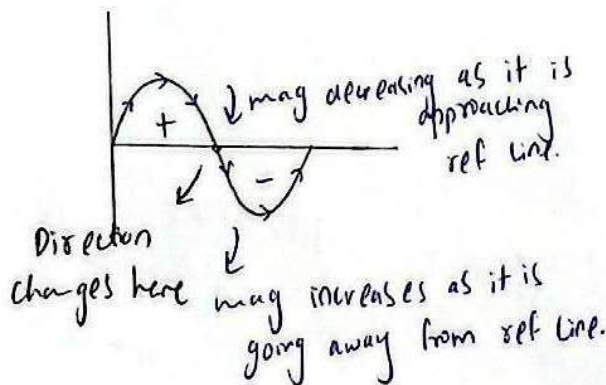
$$I = \sqrt{2} I_{rms} \sin \omega t$$

$$V = V_0 \sin \omega t$$

$$V = \sqrt{2} V_{rms} \sin \omega t$$

$f = 50Hz$ means 50 cycles/sec
In 1 cycle → 2 peaks +ve & -ve peak
↳ max time bright in 1 sec is
↳ 2f

Peak value of A.C. - highest value reached by voltage or current in one cycle.



Sinusoidal A.C

↳ magnitude
↳ continuously changes.

↳ Direction
↳ periodically changes.

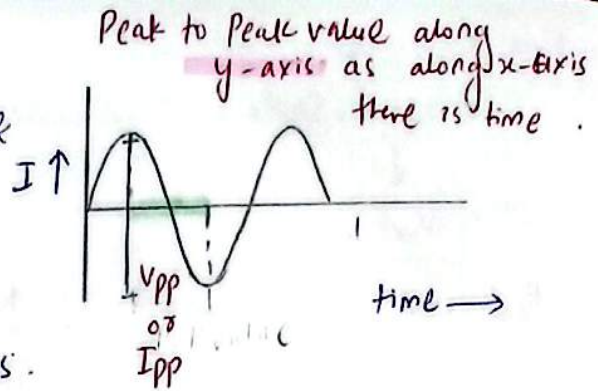
In one cycle → direction changes one time

↓
"f" time direction changes per second.
↳ frequency.

Peak to Peak value of A.C.

"a sum of consecutive positive & negative peak values".

while calculating peak to peak value we ignore negative signs.



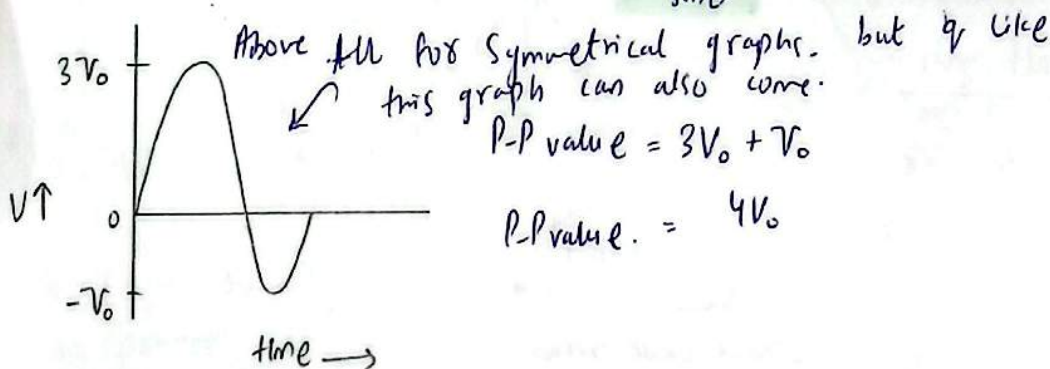
$$P-P \text{ value} = I_0 + I_0 = 2 I_0 = 2 \sqrt{2} I_{rms}$$

$$P-P \text{ value} = V_0 + V_0 = 2 V_0 = 2 \sqrt{2} V_{rms}$$

$$\therefore I = I_0 \sin \omega t, \quad I = I_0 \sin \theta$$

$$I_0 = \frac{I}{\sin \theta}$$

$$P-P \text{ value} = 2 I_0 = \frac{2 I}{\sin \theta}$$



↳ Main reason for world wide use of A.C is that it can be transmitted to a long distance easily at low cost.

$$\begin{aligned} I_1 &= 100 \sin 100\pi t \\ I_2 &= 100 \sin 200\pi t \end{aligned} \quad \left. \begin{array}{l} \\ \end{array} \right\} \text{the relationship between frequencies of AC}$$

$$\boxed{f_2 = 2f_1}$$

Houses

(220V, 50Hz)

$\downarrow \quad \downarrow$
 $V_{rms} \quad f$

Peak value = ? $V_{rms} = \frac{V_0}{\sqrt{2}}$

$$V_0 = \sqrt{2} (220)$$

$$V_0 \approx 311V$$

For full cycle,

$$\langle \sin \omega t \rangle = \overline{\sin \omega t} = 0$$

$$\langle \cos \omega t \rangle = \overline{\cos \omega t} = 0$$

$$\langle \sin^2 \omega t \rangle = \overline{\sin^2 \omega t} = 1/2$$

$$\langle \cos^2 \omega t \rangle = \overline{\cos^2 \omega t} = 1/2$$

For half cycle,

$$\overline{\sin \omega t} = 2/\pi$$

$$\overline{\cos \omega t} = 2/\pi$$

$$\overline{\sin^2 \omega t} = 1/2$$

$$\overline{\cos^2 \omega t} = 1/2 \quad \text{constant value mean is equal to that value}$$

$$\text{Time period} = T = \frac{t}{n} = \frac{1}{f}$$

In pakistan $f = 50 \text{ Hz}$

$$T = 1/50 \text{ Hz} = 0.02 \text{ s}$$

Note: capacitor blocks Steady State D.C

$$\text{D.C } [f=0]$$

$$X_c = \frac{1}{2\pi f C} = \infty$$

but capacitor permits pulsating D.C

$$f \neq 0 \\ X_c \neq 0$$

$$I = I_0 \sin \omega t$$

$$I_{rms} = \frac{I_0}{\sqrt{2}} \quad \text{only displacement current can pass through capacitor}$$

(OR)

$$I_{rms} = \sqrt{\overline{I^2}}$$

$$\begin{matrix} \textcircled{3} & \downarrow & \textcircled{1} \\ & \textcircled{2} & \end{matrix}$$

$$= \sqrt{\overline{I_0^2 \sin^2 \omega t}}$$

$$= \sqrt{\overline{I_0^2}} \sqrt{\overline{\sin^2 \omega t}}$$

$$= \sqrt{I_0^2} \sqrt{1/2}$$

$$= \frac{I_0}{\sqrt{2}}$$

$$I = \underbrace{I_0}_a + \underbrace{I_0 \sin \omega t}_b$$

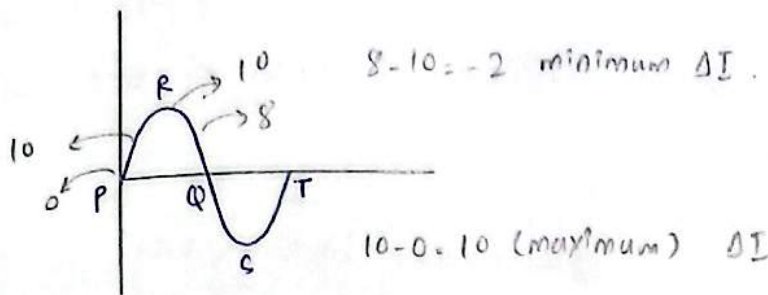
$$I_{rms} = ? \quad I_{rms} = \sqrt{\overline{I^2}}$$

$$I_{rms} = \sqrt{\overline{I_0^2} + \overline{I_0^2 \sin^2 \omega t} + 2 \overline{I_0 I_0 \sin \omega t}}$$

$$I_{rms} = \sqrt{I_0^2 + I_0^2 (1/2) + 2 I_0^2 (0)}$$

$$I_{rms} = \sqrt{2 I_0^2 (1/2) + I_0^2} = \sqrt{I_0^2 + I_0^2}$$

$$I_{rms} = \sqrt{\frac{2 I_0^2 + I_0^2}{2}} = \sqrt{\frac{3 I_0^2}{2}} = I_0 \sqrt{3/2}$$



↳ At P, Q, T value of $I = 0$ but ΔI & ΔV is maximum, difference is maximum.

↳ At R, S both values I & V are non-zero but ΔI & ΔV is minimum.

Slope of graph:

angle of tangent with the x -axis is θ $\rightarrow$ angle of tangent with the x -axis is zero.
Slope decreases & then reaches zero.

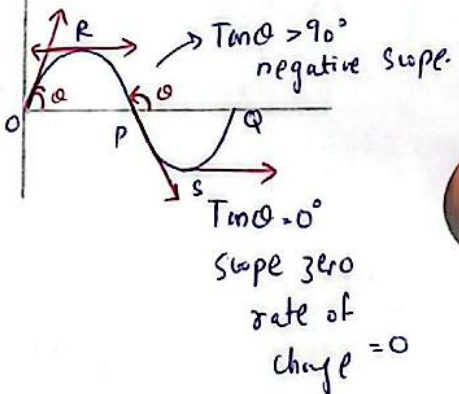
At 0 $\theta < 90^\circ$

Slope would be positive.

At P $\theta > 90^\circ$

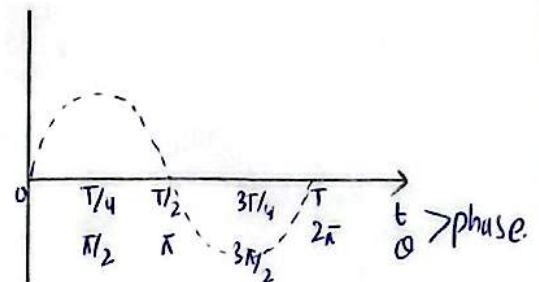
Slope would be negative.

Slope +ve if $\tan \theta < 90^\circ$



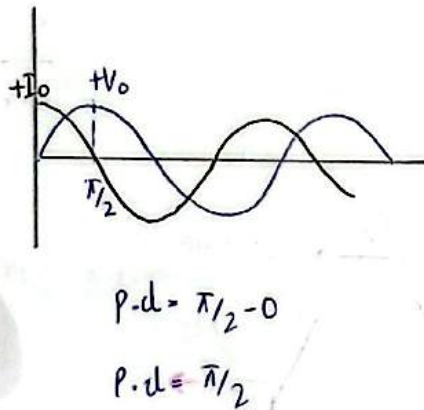
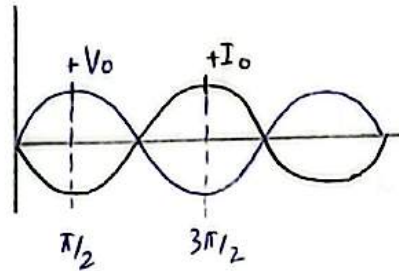
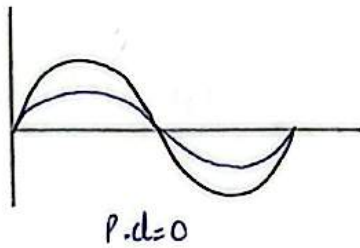
Phase & Phase difference:

Phase:- The angle or time which specifies the instantaneous value of alternating voltage or current.
 ω rad $\rightarrow$ t sec



phase difference :- reference point $\rightarrow$ the (positive) peak value

Phase angle which describes the difference of phases of two alternating quantities.



Power dissipation in resistor for Alternating current:

$P = I^2 R$ mean power is
 $P = I_0^2 \sin^2 \omega t R$ half of maximum power

$\langle P \rangle = \frac{I_0^2 R}{2}$

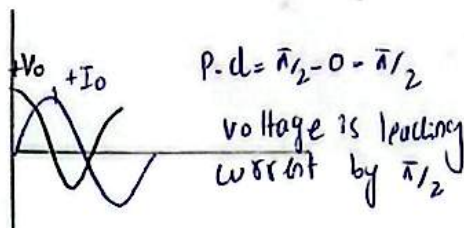
$\langle P \rangle = P_0/2$

$\langle P \rangle = \frac{I_0}{\sqrt{2}} \cdot \frac{I_0}{\sqrt{2}} R$

$\langle P \rangle = I_{rms}^2 R$ $\frac{I_0}{\sqrt{2}} = I_{rms}$

Phase lead :-

Phase difference of a quantity is said to be phase leading if the phase angle ϕ is greater than that of some quantity.

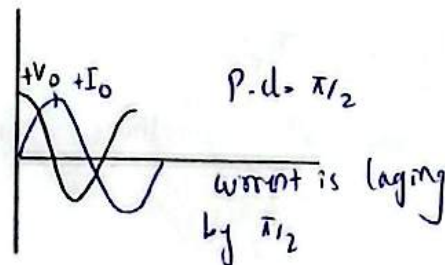


$V = V_0 \sin(\omega t + \pi/2)$

$I = I_0 \sin \omega t$

Phase lag :-

Phase difference of a quantity is said to be phase lagging if the phase angle is lesser than that of some quantity.

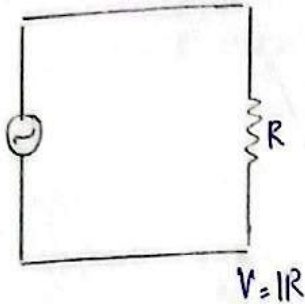


$V = V_0 \sin \omega t$

$I = I_0 \sin(\omega t - \pi/2)$

$\sin(\omega t - \pi/2) \rightarrow -\cos \omega t$
reason that current lags.

AC through Resistor



V & I are in phase
P.f. = 1

In houses voltage peak is greater than current peak & length of phasor shows peak value.

$$V = V_0 \sin \omega t$$

$$I = I_0 \sin \omega t$$

Average power dissipation :-

$$\langle P \rangle = I_{rms} V_{rms} \cos \theta$$

$$\langle P \rangle_{max} = I_{rms} V_{rms}$$

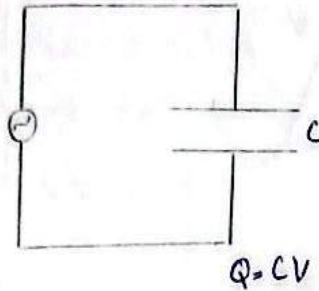
(or)

$$\langle P \rangle = \frac{I_0 V_0}{2} \cos \theta$$

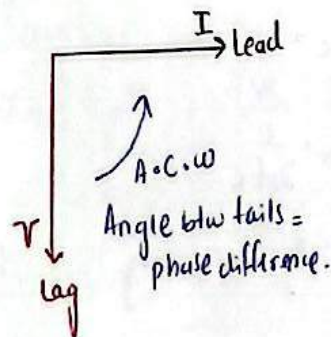
$$P_{max} = \frac{I_0 V_0}{2}$$

Resistor power dissipation is maximum.

AC through capacitor



Current is leading voltage
by $\pi/2$
P.f. = 0



$$V = V_0 \sin \omega t$$

$$I = I_0 \cos \omega t$$

$$I = I_0 \sin(\pi/2 + \omega t)$$

Average Power dissipation :-

$$\langle P \rangle = I_{rms} V_{rms} \cos \theta$$

$$\because \cos 90^\circ = 0$$

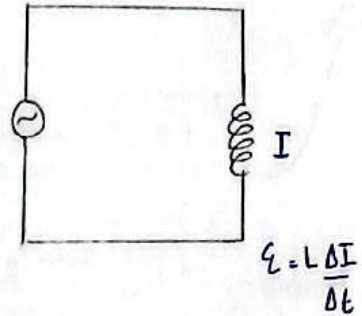
$$\langle P \rangle = 0$$

capacitor $\Rightarrow$ energy storing device. In one cycle
charge 2 times discharge 2 times.

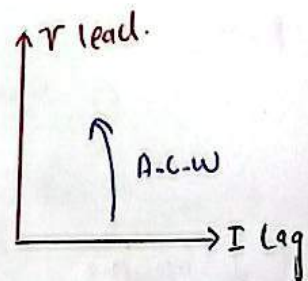
$$\langle P \rangle = 0$$

Power dissipation is minimum

AC through Inductor



voltage is leading current
by $\pi/2$
P.f. = 0



$$V = V_0 \cos \omega t$$

$$I = I_0 \sin(\pi/2 + \omega t)$$

$$I = I_0 \sin \omega t$$

Average power dissipation :-

$$\langle P \rangle = I_{rms} V_{rms} \cos \theta$$

$$\because \cos 90^\circ = 0$$

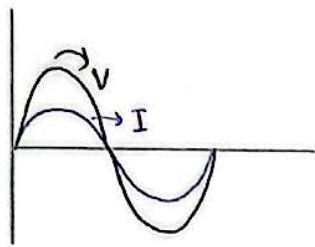
$$\langle P \rangle = 0$$

Inductor ~~capacitor~~ is also energy storing device. In one cycle
charge 2 times discharge 2 times

$$\langle P \rangle = 0$$

Power dissipation is minimum.

AC through Resistor



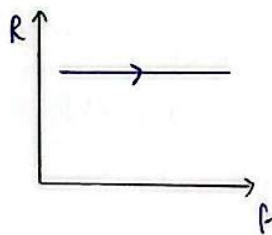
Power factor,

↓
no unit $\cos \theta = \cos 0^\circ = 1$

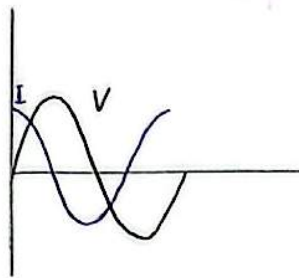
Resistance,

$$R = \frac{V}{I} \Omega$$

Effect of frequency,



AC through capacitor



Power factor,

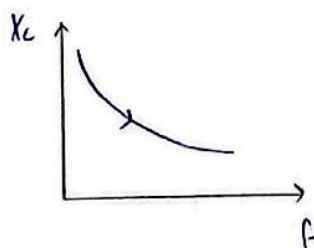
$$\cos \theta = \cos 90^\circ = 0$$

capacitive Reactance,

$$X_C = \frac{1}{\omega C} \Omega$$

$$X_C = \frac{1}{2\pi f C}$$

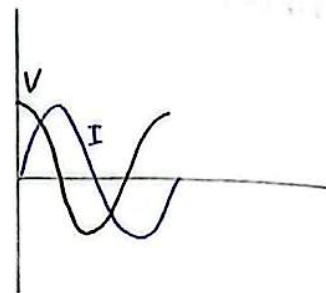
Effect of frequency,



$$I_0 = C V_0 \omega$$

~~Handwritten scribbles~~

AC through Inductor



Power factor,

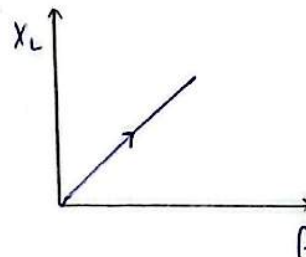
$$\cos \theta = \cos 90^\circ = 0$$

Inductive Reactance,

$$X_L = \omega L \Omega$$

$$X_L = 2\pi f L$$

Effect of frequency,



$$V_0 = L I_0 \omega$$

~~Handwritten scribbles~~

$$\overbrace{(\text{vector})}^{\omega=0^\circ} = (\text{vector})(\text{scalar})$$

$$\overbrace{(\text{phasor})}^{\omega=0^\circ} = (\text{phasor})(\text{scalar})$$

But angle between current & voltage is 90° so this rule won't be applicable here, so we will take constant values of V & I i.e.

$$V_{\text{rms}} \text{ or } V_0 \quad \bigg| \quad I_{\text{rms}} \text{ or } I_0$$

$$\left. \begin{matrix} V_{\text{rms}} \\ I_0 \end{matrix} \right\} \text{constant values.}$$

impedance :- combined effect of resistance & reactance is called impedance.

$$RC \Rightarrow Z = \sqrt{R^2 + X_C^2}$$

$$RL \Rightarrow Z = \sqrt{R^2 + X_L^2}$$

$$RLC_{\text{series}} \Rightarrow Z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$RL \rightarrow V = I \sqrt{R^2 + X_L^2}$$

$$\phi = \tan^{-1} \left(\frac{2\pi f L}{R} \right)$$

$$\boxed{\tan \phi = \frac{X_L}{R}}$$

$$\text{current unit} \Rightarrow \frac{\text{coulomb}}{\text{sec}} = \text{Ampere.}$$

$$V_{\text{net}} = \sqrt{V_R^2 + V_L^2}$$

↳ formula valid for

Peak value

RMS value

Instantaneous value

$$RC \rightarrow V = \sqrt{R^2 + X_C^2}$$

$$\phi = \tan^{-1} \left(\frac{-X_C}{R} \right)$$

$$\phi = \tan^{-1} \left(-\frac{1}{2\pi f C} \right) \left(\frac{1}{R} \right)$$

-ve sign shows that current leads & voltage lags.

$$V_{\text{net}} = \sqrt{V_R^2 + V_C^2}$$

↳ formula valid for

Peak value

RMS value

Instantaneous value.

$$\text{Real power } P = V_{\text{rms}} I_{\text{rms}} \cos \phi$$

$$P = \frac{V_0}{\sqrt{2}} \frac{I_0}{\sqrt{2}} \cos \phi$$

$$P = \frac{V_0 I_0 \cos \phi}{2}$$

$$\text{Apparent Power } P = VI$$

$$\text{Power factor } \frac{P_{\text{real}}}{P_{\text{app}}}$$

↓
ratio of real power to apparent power.

Power factor represents power loss

BEAUTIFUL CONCEPTS:

In one complete cycle / revolution of A.C $\rightarrow$ ^{or voltage} current changes its direction **2 times.**

AC $f = 50 \text{ Hz}$
 $\downarrow$
 Household $\text{So } \frac{\text{waves}}{\text{second.}}$

100 times current changes its direction.

$$\begin{aligned} i_{\text{avg}} &= 0 \text{ full cycle} \\ i_{\text{avg}} &= \frac{2i_0}{\pi} \text{ half cycle.} \\ i_{\text{rms}} &= \frac{i_0}{\sqrt{2}} \end{aligned}$$

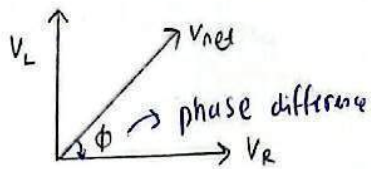
$$\begin{aligned} V_{\text{avg}} &= 0 \text{ full cycle} \\ V_{\text{avg}} &= \frac{2V_0}{\pi} \text{ half cycle} \\ V_{\text{rms}} &= \frac{V_0}{\sqrt{2}} \end{aligned}$$

Note :- I_0 & I_{rms} are values whereas I is expression.

$$\begin{aligned} I_0 &= \frac{V_0}{X_L} \checkmark, \quad I_{\text{rms}} = \frac{V_{\text{rms}}}{X_L} \checkmark, \quad I = \frac{V}{X_L} \times \\ &\quad \swarrow \text{for peak values} \quad \searrow \text{for instantaneous its wrong} \\ &\quad \text{there is not any phase exist.} \quad \downarrow \text{as current has different phase than voltage.} \end{aligned}$$

$$I_0 = \frac{V_0}{X_C} \checkmark, \quad I_{\text{rms}} = \frac{V_{\text{rms}}}{X_C} \checkmark, \quad I = \frac{V}{X_C} \times$$

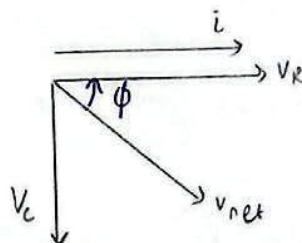
In Series R-L circuit:



due to inductor $\rightarrow$ current lags behind by ϕ not by 90° that was in pure inductor.

$$\begin{aligned} V &= V_0 \sin \omega t \\ I &= I_0 \sin(\omega t - \phi) \\ I_0 &= \frac{V_0}{Z}, \quad I_{\text{rms}} = \frac{V_{\text{rms}}}{Z} \end{aligned}$$

In Series R-C circuit:

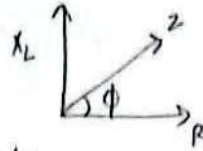


due to capacitor, current leads forward by ϕ not by 90° that was in pure capacitor.

$$\begin{aligned} V &= V_0 \sin \omega t \\ I &= I_0 \sin(\omega t + \phi) \\ I_0 &= \frac{V_0}{Z}, \quad I_{\text{rms}} = \frac{V_{\text{rms}}}{Z} \end{aligned}$$

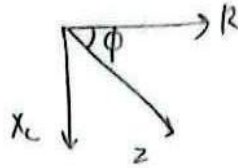
Power factor in L-R circuit.

$$\cos \phi = \frac{R}{Z}$$



Power factor in R-C circuit.

$$\cos \phi = \frac{R}{Z}$$



Electromagnetic WAVES:

↳ medium independent.

↳ oscillating $\vec{E}$ & $\vec{B}$ fields

↳ $\vec{E}$ is perpendicular to $\vec{B}$

↳ vacuum $\rightarrow$ Speed of light.

velocity of electromagnetic wave $= V = \vec{E} \times \vec{B}$

Maxwell's Equation & Prediction.

$\Delta \phi_m$ produce Electric field

$\Delta \phi_e$ produce Magnetic field.

) Em wave.

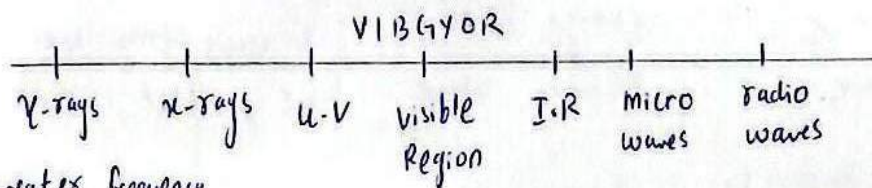
Maxwell Equation based on 3 laws.

↳ Gauss's Law
(electric flux)

↳ Ampere's Law

↳ FARADAY'S LAW.

Free Space (vacuum) $\cdot c = \frac{1}{\sqrt{\epsilon_0 \mu_0}} = 3 \times 10^8 \text{ m/s} \quad c = f\lambda$



FM has greater frequency than Am.

For long range $\rightarrow$ Am
Short range $\rightarrow$ FM

$f \downarrow, \lambda \uparrow, E \downarrow$

Am $\rightarrow$ Amplitude modulation

FM $\rightarrow$ Frequency modulation.

AM $\longrightarrow$ 530 kHz - 1170 kHz $\lambda \uparrow, f \downarrow$
FM $\longrightarrow$ 88 MHz - 108 MHz $\lambda \downarrow, f \uparrow$

→ Due to greater wavelength of A.M it is for long range.

→ Due to greater frequency of F-M it has better sound quality than A-M.

EMW $\rightarrow$ generated by transmission antenna $\rightarrow$ oscillating charges $\Delta\phi_e$ & $\Delta\phi_m$
 $\underbrace{\hspace{10em}}_{\text{Produce } \vec{E} \text{ \& \> } \vec{B} \text{ field.}}$

Transmission of electromagnetic waves requires modulation. it is a phenomenon in which we change our signals.

carrier wave ↙ high frequency + low frequency
EMW signals.

↓
Signal that is
to be transmitted without
modulation it won't transmit.

modulation has two types

Am

Amplitude modulation

change occurs in
carrier wave

FM

Frequency modulation.

change obs in
cassio's wand.

Tuning:- $f = f_{LC}$ $\rightarrow$ Circuit frequency.

Reception of Em waves

Principal used for reception of
EMW $\Rightarrow$ resonance phenomenon.

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

To regain EMW we use LC parallel circuit.

f_r = resonance frequency.

Inductance, Tendency of an circuit component to oppose a change in the electric current flowing through it.

→ it is the property of an electrical conductor (like a coil of wire) that describes its ability to store energy in a magnetic field when electric current flows through it.

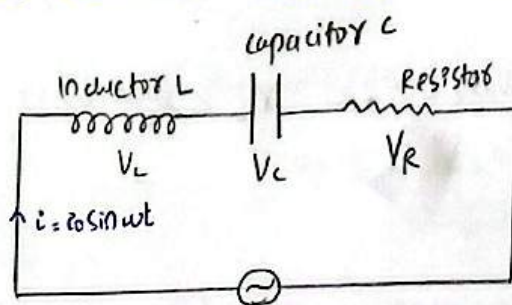
Unit VmA^{-1} or Henry.

$$L \propto \mu \frac{N^2 A}{l}$$

$$L \propto \mu$$

Note :: In RLC Series circuit the true condition of resonance takes when $X_L = X_C$.

→ LCR Series circuit :- also known as acceptor circuit.



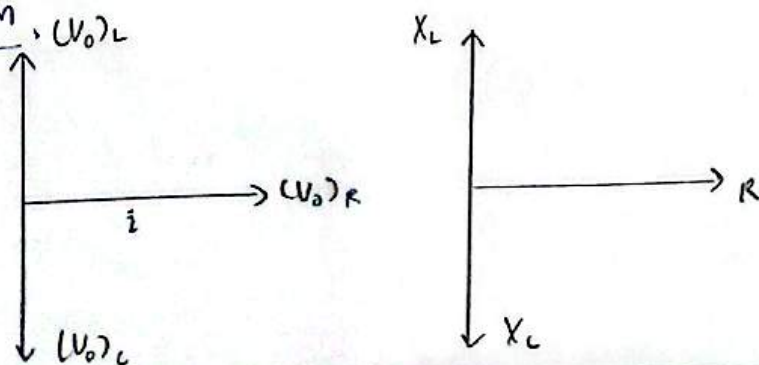
$$V_R = (V_0)_R \sin \omega t \quad \therefore (V_0)_R = I_0 R$$

$$V_L = (V_0)_L \sin (\omega t + \pi/2) \quad \therefore (V_0)_L = I_0 X_L$$

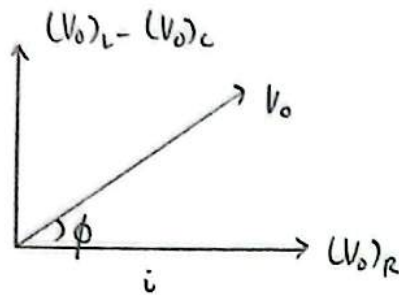
$$V_C = (V_0)_C \sin (\omega t - \pi/2) \quad \therefore (V_0)_C = I_0 X_C$$

→ you can either take V_0 or V_{rms} not V

Phasor diagram

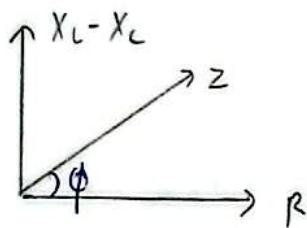


CASE # (01) when $(V_o)_L > (V_o)_C$ or $X_L > X_C$



$$I = I_o \sin \omega t$$

$$V = V_o \sin(\omega t + \phi)$$



$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$\cos \phi = \frac{R}{Z}$$

$$(V_o)^2 = (V_o)_R^2 + [(V_o)_L - (V_o)_C]^2$$

$$(V_o)^2 = I_o^2 R^2 + [I_o X_L - I_o X_C]^2$$

$$(V_o)^2 = I_o^2 R^2 + I_o^2 [X_L - X_C]^2$$

$$V_o^2 = I_o^2 [R^2 + (X_L - X_C)^2]$$

$$V_o = I_o \sqrt{R^2 + (X_L - X_C)^2}$$

impedance

$$V_o = I_o Z$$

$$\tan \phi = \frac{X_L - X_C}{R}$$

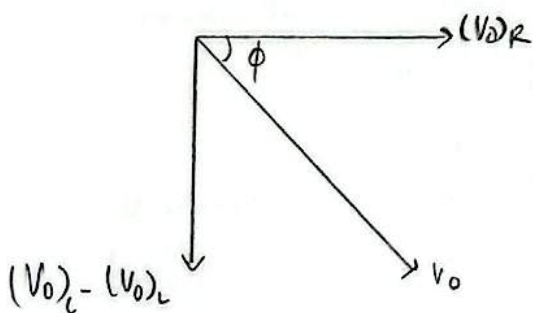
$$\phi = \tan^{-1} \left(\frac{X_L - X_C}{R} \right)$$

CASE # (02)

when $(V_o)_L < (V_o)_C$ or $X_L < X_C$

$$I_o X_L < I_o X_C$$

$$X_L < X_C$$



$$I = I_o \sin \omega t$$

$$V = V_o \sin(\omega t - \phi)$$

$$(V_o)^2 = (V_o)_R^2 + [(V_o)_C - (V_o)_L]^2$$

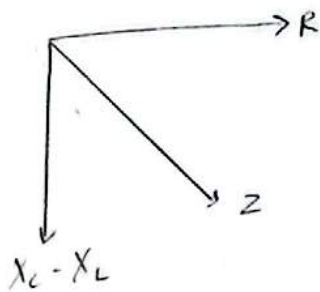
$$V_o^2 = I_o^2 R^2 + [I_o X_C - I_o X_L]^2$$

$$V_o^2 = I_o^2 R^2 + I_o^2 [X_C - X_L]^2$$

$$V_o^2 = I_o^2 [R^2 + (X_C - X_L)^2]$$

$$V_o = I_o \sqrt{R^2 + (X_C - X_L)^2}$$

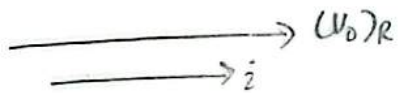
$$V_o = I_o Z$$



$$Z = \sqrt{R^2 + (X_C - X_L)^2}$$

$$\phi = \tan^{-1} \left(\frac{X_C - X_L}{R} \right)$$

CASE # (3) $(V_0)_L = (V_0)_C$ or $X_L = X_C$ Resonance condition.



$$V_0 = (V_0)_R$$

current & voltage in same phase.

Pure Resistive circuit

$$Z = R$$

X_L & X_C depends on frequency.

$$X_L = \omega L$$

$$X_C = \frac{1}{\omega C}$$

where ω is frequency of (3)

$$V = V_0 \sin \omega t$$

$$I = I_0 \sin \omega t$$

minimum value of impedance.

At resonance the impedance of the circuit is minimum.

$$\uparrow I_0 = \frac{V_0}{Z} \downarrow$$

if Z is minimum, current (I_0) is maximum.

$$(I_0)_{\max} = \frac{V_0}{R}$$

$$\tan \phi = 0 \quad \text{as } X_C - X_L \text{ or } X_L - X_C = 0$$

$$\phi = 0^\circ$$

phase difference = 0

$$X_L = X_C \text{ at Resonance.}$$

$\omega = \omega_0$ particular value of ω

$$\omega_0 L = \frac{1}{\omega_0 C}$$

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$$2\pi f_0 = \frac{1}{\sqrt{LC}}, \quad f_0 = \frac{1}{2\pi\sqrt{LC}}$$

↳ Dimension of L/CR is same as R .

$$\frac{VSA^{-1}}{S} = V/A = R$$

↳ In frequency modulation frequency of carriers waves changes with amplitude of modulating signal.

Amplitude = I_0 = Peak value.

↳ In case of D.C $I_{rms} = I$

~~For~~ ↳ The Power dissipated in a resistor for a constant potential difference V is the same as for a sinusoidal alternating potential difference or rms value V .

$$V_{rms} = \frac{V_0}{\sqrt{2}}$$

$$V = \frac{V_0}{\sqrt{2}}$$

~~multiply~~ $V_0 = \sqrt{2} V$

For Pure resistive circuit :- there is no dependency of frequency on I_{rms} . , $I_{rms} = \frac{V_{rms}}{R}$

no frequency dependence.

For Pure Inductive circuit :-

$$X_L = \omega L \quad \& \quad X_L = 2\pi f L$$

$$X_L \propto f$$

$$I_{rms} = \frac{V_{rms}}{X_L} \quad , \quad I_{rms} \propto \frac{1}{X_L}$$

$$\boxed{I_{rms} \propto 1/f}$$

$$\text{or } \boxed{I_{rms} \propto 1/\omega}$$

For pure conductive circuit -

$$X_c = \frac{1}{\omega C}$$

$$X_c = \frac{1}{2\pi f C}$$

$$X_c \propto 1/f$$

$$I_{rms} = \frac{V_{rms}}{X_c}$$

$$I_{rms} \propto 1/X_c$$

$$I_{rms} \propto f$$

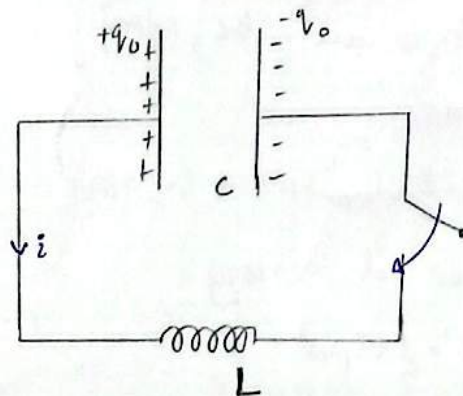
$$I_{rms} \propto \omega$$

Note -

$$I_{rms} = \frac{V_{rms}}{X_c} = \omega C V_{rms}$$

LC - OSCILLATIONS: ideal pure L-C

circuit.



Energy is oscillating between Electric field of capacitor & magnetic field of inductor.

$$E_c = \frac{q^2}{2C}$$

$$E_L = \frac{Li^2}{2}$$

if capacitor has all the energy then charge should be maximum

if inductor has all the energy then current should be maximum.

if there is no loss of energy then total energy must be constant.

$$E = \frac{1}{2} \frac{q^2}{C} = \frac{1}{2} LI^2 = \frac{1}{2} \frac{q^2}{C} + \frac{1}{2} LI^2$$

charge on each plate of capacitor also oscillates from

$$+q_0 \rightarrow 0 \rightarrow -q_0 \rightarrow 0 \rightarrow +q_0.$$

~~Kirchoff's loop law:~~

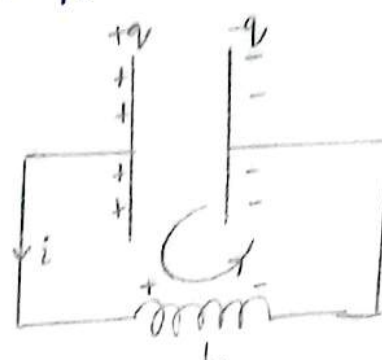
~~$$+\frac{q}{C} - L \frac{di}{dt} = 0$$~~

~~$$I = \frac{\Delta Q}{\Delta t}$$~~

~~Induced EMF:~~

~~$$I = -\frac{dQ}{dt}$$~~

Initially, the capacitor is fully charged. So initial energy stored in capacitor is $U = \frac{1}{2} CV^2$



when connected to a pure inductor, energy keeps transferring between capacitor & inductor. At the instant of maximum current, capacitor becomes completely discharged, so all the energy is magnetic energy in inductor.

$$U = \frac{1}{2} LI_{\max}^2$$

By conservation of energy:

$$\frac{1}{2} CV^2 = \frac{1}{2} LI_{\max}^2$$

$$CV^2 = LI_{\max}^2$$

$$I_{\max}^2 = \frac{CV^2}{L}$$

$$I_{\max} = V \sqrt{\frac{C}{L}}$$